Nirma University

Institute of Technology

Department of Computer Science & Engineering

Explainable AI

Lab 6: What-If Tool for Image Classification

(Interactive Exploratory Explainability with CelebA Smile Detection)

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1 Objective

The objective of this lab is to demonstrate the use of Google's **What-If Tool (WIT)** — an interactive visual interface for exploring machine learning model behavior. The tool is applied to a **smile detection CNN** trained on a subset of the CelebA dataset, enabling analysis of model performance, fairness, and counterfactual reasoning.

2 Dataset and Model

2.1 CelebA Dataset

- **Full Dataset:** CelebA (CelebFaces Attributes Dataset) contains over 200,000 celebrity face images with 40 binary attribute annotations
- **Subset Used:** A test subset of 250 images with associated metadata
- **Task:** Binary classification — Smiling vs. Not Smiling
- **Image Resolution:** $78 \times 64 \times 3$ (resized face crops)
- **Metadata Attributes:** Gender, Young/Not Young, smiling ground truth, attractiveness scores, and other facial attributes

2.2 CNN Architecture

The pre-trained model is a custom Convolutional Neural Network built with TensorFlow/Keras:

Table 1: CNN architecture for smile detection

Layer	Output Shape	Parameters
Input	$(78, 64, 3)$	—
Conv2D (32 filters, 5×5, ReLU)	$(74, 60, 32)$	2,432
MaxPooling2D (2×2)	$(37, 30, 32)$	—
Conv2D (64 filters, 5×5, ReLU)	$(33, 26, 64)$	51,264
MaxPooling2D (2×2)	$(16, 13, 64)$	—
Flatten	$(13, 312)$	—
Dense (256, ReLU)	(256)	3,408,128
Dense (2, Softmax)	(2)	514

The model outputs softmax probabilities for two classes: `[Not Smiling, Smiling]`.

3 Methodology

3.1 Data Preparation

3.1.1 Protobuf Compatibility

A patch was applied to the `google.protobuf` library to handle deprecation of the `including_default_v` parameter in newer versions. This ensures compatibility between the WIT widget and the current TensorFlow/protobuf ecosystem.

3.1.2 Data Conversion

The dataset was converted from CSV + image files to TensorFlow `tf.Example` protobuf format:

1. Each image was loaded from disk, converted to PNG bytes, and stored in the `image/encoded` feature field
2. Metadata attributes (gender, age, smiling label, etc.) were stored as additional feature fields
3. This unified format enables the What-If Tool to access both image data and metadata for analysis

3.2 Custom Prediction Function

A custom prediction function was implemented to bridge the WIT widget with the Keras model:

1. Extract `image/encoded` bytes from each `tf.Example`
2. Decode bytes to PIL Images using `BytesIO`
3. Convert images to numpy arrays and normalize to $[0.0, 1.0]$ range
4. Run batch inference through the CNN
5. Return softmax probabilities as a numpy array of shape $(n, 2)$

3.3 What-If Tool Configuration

1. Selected 250 test examples for visualization
2. Configured `WitConfigBuilder` with the custom prediction function
3. Set label vocabulary to `["Not Smiling", "Smiling"]`
4. Launched the interactive widget with a height of 700 pixels

4 What-If Tool: Features and Analysis

4.1 Overview of WIT Capabilities

The What-If Tool provides several interactive panels for model exploration:

- **Datapoint Editor:** Inspect and modify individual test examples
- **Performance Tab:** Analyze model accuracy across data slices
- **Features Tab:** Visualize feature distributions and partial dependence
- **Fairness Tab:** Compare model performance across demographic groups

4.2 Performance Analysis

With the ground truth feature set to "Smiling":

- The tool computes overall accuracy, precision, recall, and F1-score
- Scatter plots of "inference correct" vs. other features reveal performance disparities
- Binning by features like "Young" or "Male" shows whether accuracy varies across subgroups

4.3 Counterfactual Analysis

The "nearest counterfactual" feature is one of WIT's most powerful capabilities:

- For a selected image, the tool finds the most similar image (measured by average pixel color) that has a **different** prediction
- This enables reasoning like: "What minimal change would flip the model's decision from Smiling to Not Smiling?"
- Counterfactual analysis helps identify potential failure modes and spurious correlations

4.4 Fairness Exploration

- Compare model accuracy between Male and Female subgroups
- Investigate whether the smile detector performs equally well on different age groups
- Identify potential biases: Does the model associate certain non-smile attributes (e.g., attractiveness) with smiling?

4.5 Important Considerations

- This model uses **only image pixels** as input. Therefore, partial dependence plots for metadata features (age, gender) are **flat** — these features do not influence predictions
- The metadata attributes are used exclusively for **slicing and analysis**, not for prediction
- The WIT is an interactive tool — visualizations are rendered at runtime and cannot be captured as static images

5 Discussion

5.1 WIT vs. Algorithmic XAI Methods

Table 2: Comparison of What-If Tool with other XAI methods

Property	What-If Tool	SHAP/LIME/Grad-CAM
Interaction type	Interactive visual	Programmatic
Explanation level	Dataset-wide + individual	Individual or global
Fairness analysis	Built-in	Requires additional code
Counterfactuals	Native support	Not standard
Model-agnostic	Yes	Varies
Technical expertise	Low (visual interface)	High (coding required)

5.2 Advantages of Interactive Explainability

1. **Accessibility:** Non-technical stakeholders (product managers, compliance officers) can explore model behavior without coding
2. **Discovery:** Interactive exploration often surfaces unexpected patterns that predefined analyses miss
3. **Fairness auditing:** Built-in tools for demographic analysis facilitate responsible AI practices
4. **Counterfactual reasoning:** "What would need to change for a different outcome?" is a natural human reasoning pattern that WIT supports natively

5.3 Limitations

- WIT is primarily designed for Google's ecosystem (TensorFlow models, Colab/Jupyter)
- Scalability is limited — exploring millions of examples interactively is not feasible
- The counterfactual metric (average pixel color) is simplistic; more sophisticated similarity measures could yield more meaningful counterfactuals

6 Conclusion

This lab demonstrated that the What-If Tool provides a complementary approach to model explainability:

- It enables **interactive, visual exploration** of model behavior without writing code
- It bridges the gap between technical model analysis and stakeholder communication
- The counterfactual analysis and fairness auditing features are particularly valuable for deploying AI models responsibly
- When combined with algorithmic methods (SHAP, LIME, Grad-CAM), WIT provides a comprehensive framework for understanding, debugging, and trusting machine learning models

The What-If Tool represents an important evolution in XAI — moving from static explanations to dynamic, user-driven exploration of model behavior.